

Online learning platform

(A) Student Functional Requirement

1. Account and Profile Management

FR-S01: The system shall allow students to register and create an account on the Online Learning Portal.

FR-S02: The system shall allow registered students to securely log in, log out, and reset their password.

FR-S03: The system shall allow students to view and update their profile information.

2. Course Discovery and Enrollment

FR-S05: The system shall allow students to browse and search for available courses.

FR-S06: The system shall allow students to filter courses based on criteria such as domain, duration, rating, and subject.

FR-S07: The system shall allow students to view detailed course information, including course description, syllabus, instructor, prerequisites, duration, and learning outcomes.

FR-S08: The system shall allow students to enroll in courses available to them.

FR-S09: The system shall allow students to view their enrolled courses.

FR-S10: The system shall allow students to bookmark courses for future reference.

FR-S11: The system shall allow registered students to download and view their earned certificate.

3. Learning and Course Content

FR-S11: The system shall allow enrolled students to access the learning content of their courses.

FR-S12: The system shall allow students to watch, pause, resume, and seek through video lectures.

FR-S13: The system shall remember the student's video playback position and allow them to resume from where they stopped.

FR-S14: The system shall allow students to access and download permitted learning materials such as notes, PDFs, presentations, and other resources.

FR-S15: The system shall allow students to mark learning content as completed.

FR-S16: The system shall track and display the student's progress within each enrolled course.

FR-S17: The system shall allow students to bookmark important lectures and learning resources.

FR-S18: The system shall allow students to search for specific topics or keywords within permitted video lectures and access the relevant portion of the lecture.

4. Quiz Management

FR-S18: The system shall allow students to view and attempt quizzes associated with their enrolled courses.

FR-S19: The system shall display quiz instructions, duration, number of questions, and maximum marks.

FR-S20: The system shall support different types of quiz questions, including multiple-choice, multiple-select, true/false, numerical, and descriptive questions where configured.

FR-S21: The system shall save the user's response after it clicks the save and next option.

FR-S22: The system shall display the remaining time for timed quizzes.

FR-S23: The system shall allow students to navigate between questions and submit the quiz.

FR-S24: The system shall automatically submit the quiz when the allotted time expires.

FR-S25: The system shall allow students to view their quiz marks and results after submission, according to the configured assessment settings.

FR-S26: The system shall allow students to review their answers and feedback when enabled.

6. Examination Management

FR-S33: The system shall allow students to view upcoming, ongoing, and completed examinations.

FR-S34: The system shall display examination instructions, schedule, duration, maximum marks, and eligibility requirements.

FR-S35: The system shall allow eligible students to start an examination during the permitted examination period.

FR-S36: The system shall automatically save examination responses during an active examination.

FR-S37: The system shall display the remaining examination time and automatically submit the examination when the allotted time expires.

FR-S38: The system shall allow students to submit their examination manually.

FR-S39: The system shall allow students to view their examination results and marks after evaluation and publication.

FR-S40: The system shall allow students to view question-wise marks and feedback when permitted.

7. Question Reporting and Ambiguity

FR-S41: The system shall allow students to report ambiguous, incorrect, incomplete, or otherwise problematic questions in quizzes and examinations.

FR-S42: The system shall allow students to select a reason for reporting a question and provide an explanation.

FR-S43: The system shall allow students to view the status of their submitted question reports.

FR-S44: The system shall notify students about the resolution of their reported questions.

FR-S45: The system shall apply instructor-approved corrections or question invalidations to affected assessment results according to the configured assessment policy.

8. Marks and Performance Tracking

FR-S46: The system shall allow students to view marks obtained in quizzes, assignments, and examinations.

FR-S47: The system shall display the student's overall performance and course completion percentage.

FR-S48: The system shall maintain a history of the student's assessment results.

FR-S49: The system shall provide topic-wise performance information where supported.

FR-S50: The system shall identify topics in which students require additional practice based on their performance.

FR-S51: The system shall allow students to track their academic performance over time.

FR-S52: The system shall allow students to create or receive a personalized study schedule based on their courses, deadlines, and learning progress.

9. Examination Monitoring and Anti-Cheating

FR-S52: The system shall maintain an activity log for each monitored examination attempt.

FR-S53: The system shall detect and record when a student leaves the examination browser tab or window during a monitored examination.

FR-S54: The system shall record the timestamp and duration of each detected focus-loss or tab-change event.

FR-S55: The system shall notify students when configured examination-monitoring rules are triggered.

FR-S56: The system shall support randomized question and answer-option ordering during examinations. **FR-S57:** The system shall support generating different examination sets from a configured question bank.

FR-S58: The system shall support fullscreen examination mode where supported by the browser.

FR-S59: The system shall request explicit permission from students before accessing screen-sharing functionality.

FR-S60: The system shall record when screen sharing starts, stops, or is unexpectedly terminated during a monitored examination.

FR-S61: The system shall associate examination-monitoring events with the corresponding student's examination attempt.

10. Communication and Interaction

FR-S62: The system shall allow students to view course announcements.

FR-S63: The system shall allow students to ask questions related to course content.

FR-S64: The system shall allow students to participate in course discussions and view instructor responses.

FR-S65: The system shall notify students about important course events, deadlines, newly available content, examinations, and announcements.

11. Gamification and Engagement

FR-S66: The system shall award points to students for configured learning activities.

FR-S67: The system shall track student learning streaks.

FR-S68: The system shall award badges for predefined achievements.

FR-S69: The system shall allow students to view their achievements, badges, and points.

FR-S70: The system shall provide optional leaderboards and learning challenges where enabled.

12. AI-Based Learning Assistance

FR-S71: The system shall provide an AI-based learning assistant for course-related queries.

FR-S72: The system shall allow students to ask questions about permitted course content through the AI learning assistant.

FR-S73: The system shall generate summaries of permitted course videos and learning materials.

FR-S74: The system shall provide explanations for selected assessment questions where enabled.

FR-S75: The system shall identify weak topics based on the student's learning and assessment performance.

FR-S76: The system shall recommend relevant learning resources and practice questions based on the student's learning activity and performance.

13. Course Completion and Certificates

FR-S77: The system shall determine whether a student has satisfied the configured completion requirements of a course.

FR-S78: The system shall allow eligible students to view and obtain their course completion certificate.

FR-S79: The system shall maintain a record of certificates earned by the student.

(B) Course Instructor Functional Requirements

1. Account and Profile Management

FR-I01: The system shall allow instructors to register and create an account on the Online Learning Portal, subject to admin verification/approval.

FR-I02: The system shall allow registered instructors to securely log in, log out, and reset their password.

FR-I03: The system shall allow instructors to view and update their profile information, including qualifications, expertise areas, and bio. FR-I04: The system shall allow instructors to upload and update a profile picture and supporting credentials/documents.

2. Course Creation and Management

FR-I05: The system shall allow instructors to create a new course with title, description, category, prerequisites, and learning outcomes. FR-I06: The system shall allow instructors to organize course content into modules, sections, and lessons.

FR-I07: The system shall allow instructors to save a course as a draft before publishing.

FR-I08: The system shall allow instructors to submit a course for admin review and approval prior to publishing.

FR-I09: The system shall allow instructors to edit, update, or unpublish an existing course.

FR-I10: The system shall allow instructors to set course pricing, discounts, and enrollment limits where applicable.

FR-I11: The system shall allow instructors to define prerequisites required for a course.

FR-I12: The system shall allow instructors to archive or delete a course, subject to platform policy.

FR-I58: The system shall allow instructors to associate themselves with (get assigned to or select) any course available on the platform, subject to admin approval where required.

FR-I59: The system shall allow multiple instructors to be associated with the same course.

FR-I60: The system shall allow instructors to view a list of all courses they are currently associated with.

3. Content Upload and Management

FR-I13: The system shall allow instructors to upload video lectures for their courses.

FR-I14: The system shall allow instructors to upload supplementary learning materials such as notes, PDFs, presentations, and other resources.

FR-I15: The system shall notify instructors of the processing/transcoding status of uploaded video content. FR-I16: The system shall allow instructors to reorder, replace, or remove uploaded content within a course.

FR-I17: The system shall allow instructors to mark specific learning materials as downloadable or view-only.

FR-I18: The system shall allow instructors to schedule the release date/time of specific course content.

FR-I61: The system shall allow an instructor to upload video content only to the course(s) they are associated with.

FR-I62: The system shall allow students to view/stream video content uploaded by the instructor for their enrolled course.

4. Quiz and Assessment Creation

FR-I19: The system shall allow instructors to create quizzes and associate them with specific courses or modules.

FR-I20: The system shall allow instructors to configure quiz settings, including duration, number of questions, maximum marks, and number of attempts.

FR-I21: The system shall allow instructors to add different types of quiz questions, including multiple-choice, multiple-select, true/false, numerical, and descriptive questions.

FR-I22: The system shall allow instructors to define correct answers and marking schemes for objective questions.

FR-I23: The system shall allow instructors to enable or disable answer review and feedback visibility for students after quiz submission.

FR-I24: The system shall allow instructors to maintain a question bank for reuse across quizzes and examinations.

5. Examination Management

FR-I25: The system shall allow instructors to create and schedule examinations, including start time, duration, and eligibility criteria. FR-I26: The system shall allow instructors to configure examination-monitoring rules such as tab-switch detection, fullscreen mode, and screen-sharing requirements.

FR-I27: The system shall allow instructors to generate randomized examination sets from a configured question bank.

FR-I28: The system shall allow instructors to publish examination results after evaluation.

FR-I29: The system shall allow instructors to review examination-monitoring logs and flagged activity for their examinations.

6. Grading and Evaluation

FR-I30: The system shall allow instructors to view and grade student submissions for descriptive/subjective questions.

FR-I31: The system shall allow instructors to provide feedback and comments on graded submissions.

FR-I32: The system shall allow instructors to override auto-generated grades where necessary, with a recorded justification. FR-I33: The system shall notify instructors of pending ungraded submissions.

FR-I34: The system shall allow instructors to publish or withhold assessment results according to configured release settings.

7. Question Reporting and Ambiguity Resolution

FR-I35: The system shall notify instructors when a student reports an ambiguous, incorrect, or problematic question.

FR-I36: The system shall allow instructors to review reported questions and take action such as correcting, invalidating, or dismissing the report.

FR-I37: The system shall allow instructors to apply approved corrections or invalidations to affected student results.

FR-I38: The system shall notify students about the resolution status of their reported questions.

8. Student Interaction and Communication

FR-I39: The system shall allow instructors to post announcements for their enrolled students.

FR-I40: The system shall allow instructors to respond to student questions and participate in course discussion forums. FR-I41: The system shall allow instructors to send notifications regarding deadlines, newly available content, and examinations. FR-I42: The system shall allow instructors to view and moderate discussion posts within their own courses.

9. Student Performance and Analytics

FR-I43: The system shall allow instructors to view enrollment statistics for their courses.

FR-I44: The system shall allow instructors to view aggregate and individual student performance in quizzes and examinations. FR-I45: The system shall allow instructors to view course completion rates and engagement metrics.

FR-I46: The system shall allow instructors to identify commonly missed topics or questions across their student cohort. FR-I47: The system shall allow instructors to export student performance reports.

10. Earnings and Payouts

FR-I48: The system shall allow instructors to view their earnings summary based on enrollments and course sales.

FR-I49: The system shall allow instructors to view their payout history and status.

FR-I50: The system shall allow instructors to configure or update their payout account details.

11. Gamification Configuration

FR-I51: The system shall allow instructors to configure gamification rules such as points, badges, and streak criteria for their courses. FR-I52: The system shall allow instructors to enable or disable leaderboards and learning challenges for their courses.

12. AI-Based Assistance for Instructors

FR-I53: The system shall provide instructors with AI-generated suggestions for quiz questions based on course content. FR-I54: The system shall provide instructors with AI-generated summaries of student performance trends.

FR-I55: The system shall allow instructors to review and edit AI-generated content before publishing it to students.

13. Certification

FR-I56: The system shall allow instructors to define the completion criteria required for students to earn a course certificate. FR-I57: The system shall allow instructors to view a record of certificates issued for their courses.

Admin & Payment Gateway — Functional Requirements

(C) Admin Functional Requirements

1. Account and Access Management

FR-A01: The system shall allow admins to securely log in and log out with elevated access privileges.

FR-A02: The system shall allow admins to create, update, suspend, or delete student and instructor accounts.

FR-A03: The system shall allow admins to verify and approve new instructor registrations before granting course-creation access. FR-A04: The system shall allow admins to assign or revoke admin/sub-admin roles and permissions.

FR-A05: The system shall allow admins to reset user passwords or trigger account recovery on behalf of users when required.

2. Course Approval and Management

FR-A06: The system shall allow admins to review courses submitted by instructors before publishing.

FR-A07: The system shall allow admins to approve, reject, or request changes to a submitted course, with reasons provided to the instructor.

FR-A08: The system shall allow admins to unpublish, archive, or remove a live course from the platform.

FR-A09: The system shall allow admins to view a list of all courses on the platform, filterable by status (draft, pending review, published, archived).

FR-A10: The system shall allow admins to assign or reassign instructor(s) to a course where necessary.

3. Content Moderation

FR-A11: The system shall allow admins to review flagged or reported course content, discussion posts, and comments.

FR-A12: The system shall allow admins to remove or edit content that violates platform policy.

FR-A13: The system shall allow admins to view a queue of pending moderation actions prioritized by report severity/age.

FR-A14: The system shall notify the relevant instructor or student when moderation action is taken on their content.

4. Question Reporting Oversight

FR-A15: The system shall allow admins to view escalated question reports that instructors have not resolved within a configured timeframe.

FR-A16: The system shall allow admins to override or finalize a decision on a disputed question report.

5. Examination Oversight

FR-A17: The system shall allow admins to view platform-wide examination schedules and monitoring logs.

FR-A18: The system shall allow admins to review flagged examination-monitoring incidents (tab-switch, screen-share violations) across all courses.

FR-A19: The system shall allow admins to invalidate or nullify an examination attempt in case of confirmed malpractice.

6. User and Platform Analytics

FR-A20: The system shall allow admins to view platform-wide statistics, including total users, active courses, enrollments, and revenue.

FR-A21: The system shall allow admins to generate and export reports on platform usage, engagement, and growth trends. FR-A22: The system shall allow admins to view instructor performance metrics across all their courses.

7. Financial Oversight

FR-A23: The system shall allow admins to view all transaction records, including payments, refunds, and instructor payouts. FR-A24: The system shall allow admins to approve or reject refund requests that fall outside automated policy rules.

FR-A25: The system shall allow admins to configure platform-wide pricing rules, commission percentages, and discount policies. FR-A26: The system shall allow admins to initiate or halt instructor payouts in case of disputes or policy violations.

8. Communication and Notifications

FR-A27: The system shall allow admins to broadcast platform-wide announcements to all users or specific user segments. FR-A28: The system shall allow admins to configure automated notification templates (e.g., enrollment confirmation, certificate issuance).

9. System Configuration and Compliance

FR-A29: The system shall allow admins to configure platform-wide settings such as certificate templates, gamification rules, and AI-assistant availability.

FR-A30: The system shall allow admins to manage data-retention and privacy-compliance settings in accordance with applicable regulations.

FR-A31: The system shall maintain an audit log of all admin actions for accountability and compliance purposes.

FR-A32: The system shall allow admins to configure and manage third-party integrations (payment gateway, CDN, AI services).

(D) Payment Gateway (Third-Party) Functional Requirements

These define what the platform requires from and exchanges with the integrated payment gateway provider(s) (e.g., Razorpay, Stripe, PayPal).

1. Transaction Processing

FR-PG01: The payment gateway shall process student payments for course purchases and subscriptions in supported currencies. FR-PG02: The payment gateway shall support multiple payment methods, including credit/debit cards, UPI, net banking, and digital wallets.

FR-PG03: The payment gateway shall return a real-time transaction status (success, failure, pending) to the platform.

FR-PG04: The system shall update enrollment status only after receiving a confirmed success response from the payment gateway.

2. Refunds and Reversals

FR-PG05: The payment gateway shall support processing of refunds initiated by the platform, either fully or partially.

FR-PG06: The payment gateway shall notify the platform of refund status (processed, failed, pending) via callback or webhook.

3. Payouts

FR-PG07: The payment gateway shall support disbursement of instructor payouts to linked bank accounts or wallets.

FR-PG08: The payment gateway shall provide payout status updates (successful, failed, delayed) to the platform.

4. Security and Compliance

FR-PG09: The payment gateway shall handle and store sensitive payment data (card numbers, CVV) in compliance with PCI-DSS standards, without exposing raw data to the platform. FR-PG10: The payment gateway shall support tokenization of payment methods for repeat/subscription billing.

FR-PG11: The payment gateway shall provide fraud-detection signals or risk scores for flagged transactions where available.

5. Reconciliation and Reporting

FR-PG12: The payment gateway shall provide transaction logs/reports accessible to the platform for reconciliation purposes. FR-PG13: The payment gateway shall notify the platform of any chargeback or dispute raised by a student's bank/card issuer.

6. Webhooks and Integration

FR-PG14: The payment gateway shall send webhook notifications to the platform for key events (payment success, failure, refund, payout, chargeback).

FR-PG15: The system shall verify the authenticity of incoming payment gateway webhooks (e.g., via signature validation) before processing them.

(E)NFRS

1. Performance

NFR-01: Video lectures shall begin playback within 3 seconds of a student clicking play, under normal network conditions (supports FR-S12, FR-I13, FR-I62).

NFR-02: The course catalog search/filter (FR-S05, FR-S06) shall return results within 2 seconds for a catalog of up to 10,000 courses.

NFR-03: Quiz and examination auto-save (FR-S21, FR-S36) shall persist a response within 1 second of the student's action to minimize data loss risk.

NFR-04: The AI learning assistant (FR-S71–FR-S76, FR-I53–FR-I55) shall return a response to a student/instructor query within 5 seconds for standard queries.

NFR-05: Dashboard analytics for instructors and admins (FR-I43–FR-I47, FR-A20–FR-A22) shall load within 4 seconds for datasets up to 100,000 records.

2. Scalability

NFR-06: The system shall support at least 5,000 concurrent video streams without noticeable degradation in playback quality.

NFR-07: The examination module shall support at least 2,000 students taking a monitored examination simultaneously (FR-S52–FR-S61, FR-I26, FR-A17–FR-A19) without timing or auto-submit failures.

NFR-08: The system architecture shall support horizontal scaling of application servers during peak enrollment periods (e.g., course launch, exam windows) without manual reconfiguration downtime.

3. Availability and Reliability

NFR-09: The platform shall maintain an uptime of at least 99.5%, measured monthly, excluding scheduled maintenance windows. NFR-10: The examination module shall

guarantee 99.9% availability during any active, scheduled examination window, given the criticality of FR-S35–FR-S38 and FR-I25.

NFR-11: In case of a student's connectivity loss during an active examination, the system shall preserve the last auto-saved response (FR-S36) and allow resumption within the remaining time window. NFR-12: Payment transaction processing (FR-P01, FR-PG01–FR-PG04) shall have a documented failover path if the payment gateway is temporarily unreachable, preventing partial/ghost enrollments.

4. Security

NFR-13: All user authentication (FR-S02, FR-I02, FR-A01) shall use secure password hashing (e.g., bcrypt/Argon2) and support multi-factor authentication for admin accounts.

NFR-14: All data in transit between client and server shall be encrypted using TLS 1.2 or higher.

NFR-15: Sensitive payment data (card numbers, CVV) shall never be stored on platform servers; all such data shall be handled exclusively by the PCI-DSS-compliant payment gateway (FR-PG09).

NFR-16: Incoming payment gateway webhooks (FR-PG15) shall be authenticated via signature verification before being processed, to prevent spoofed payment/refund events.

NFR-17: Role-based access control (RBAC) shall enforce that students, instructors, and admins can only access functionality and data permitted for their role (e.g., FR-I61 — instructors upload only to associated courses). NFR-18: Screen-sharing and tab-monitoring data captured during examinations (FR-S53–FR-S60) shall be stored securely and access-restricted to authorized instructors/admins only.

NFR-19: The system shall maintain an immutable audit log of all admin actions (FR-A31) and critical financial transactions (FR-P07, FR-PG12) for at least 1 year, or as required by regulation.

5. Usability and Accessibility

NFR-20: The web interface shall be responsive and fully functional across desktop, tablet, and mobile screen sizes.

NFR-21: The platform shall comply with WCAG 2.1 Level AA accessibility standards to support differently-abled learners (screen-reader compatibility, keyboard navigation, captions/transcripts for video content).

NFR-22: Core student workflows (enrollment, taking a quiz/exam, viewing certificates) shall be completable within 3 clicks/screens from the relevant landing page.

NFR-23: Error messages (e.g., failed payment, quiz submission failure) shall be presented in clear, non-technical language with a suggested next action.

6. Compatibility

NFR-24: The platform shall function correctly on the latest two major versions of Chrome, Firefox, Safari, and Edge.

NFR-25: Fullscreen examination mode (FR-S58) shall degrade gracefully with a clear warning on browsers that do not support the Fullscreen API. NFR-26: Video content shall be encoded to support adaptive bitrate streaming (HLS/DASH) for compatibility across varying network speeds and devices.

7. Maintainability

NFR-27: The system shall be built using a modular architecture (e.g., separate services for course management, assessments, payments, notifications) to allow independent updates without full-system downtime. NFR-28: The codebase shall maintain a minimum of 70% automated test coverage for critical modules (authentication, payments, examination submission).

NFR-29: All third-party integrations (payment gateway, CDN, AI services — FR-A32) shall be abstracted behind internal service interfaces to allow provider replacement with minimal code change.

8. Data Privacy and Compliance

NFR-30: The system shall comply with applicable data protection regulations (e.g., India's DPDP Act, GDPR for international users) regarding collection, storage, and deletion of personal data. NFR-31: Students shall be able to request export or deletion of their personal data, with the system fulfilling such requests within a regulator-defined timeframe (typically 30 days).

NFR-32: Examination-monitoring data (webcam/screen-share, tab-switch logs) shall only be collected with explicit student consent (FR-S59) and retained only as long as necessary for dispute resolution.

NFR-33: Financial transaction logs (FR-P07, FR-PG12) shall be retained in compliance with local financial audit regulations (commonly 7 years).

9. Backup and Disaster Recovery

NFR-34: The system shall perform automated daily backups of course content, user data, and transaction records, with a recovery point objective (RPO) of no more than 24 hours.

NFR-35: In the event of a system failure, the platform shall have a recovery time objective (RTO) of no more than 4 hours for core services (login, enrollment, video playback).

(F)Domain Requirements:

ID	Requirement Domain	Domain Source	Related FR / NFR
DR-01	Assessment results, once published, shall not be modified except through a recorded correction, override, or invalidation with justification and timestamp.	Academic integrity and grade-dispute norms	FR-S45, FR-I32, FR-I37
DR-02	Each certificate shall carry a unique ID and be verifiable by a third party (e.g., an employer) through a verification link or QR code.	Certificates are credentials; they must be checkable to be credible	FR-S77–S79, FR-I56
DR-03	A certificate shall be issued only after the course's configured completion criteria are fully met.	Certification integrity	FR-S77, FR-I56
DR-04	Students under 18 shall register only with verifiable parental/guardian consent.	India's DPDP Act (children's data)	FR-S01, NFR-30
DR-05	Student academic records (marks, attempts, certificates, monitoring logs) shall be visible only to the student, their course instructors, and authorized admins.	Education data confidentiality	NFR-17
DR-06	Exam-monitoring data (tab-switch logs, screen-share) shall be collected only after explicit consent and used only for malpractice review.	Privacy law; proportionate monitoring	FR-S59, NFR-32
DR-07	A monitoring flag alone shall not automatically penalize a student; an instructor or admin must review it before any attempt is invalidated.	Due process; tab-switch detection can give false positives	FR-I29, FR-A19
DR-08	Course content shall be used only with the instructor's declared ownership or license, and copyright complaints shall be handled through a takedown process.	Copyright Act (India)	FR-A11, FR-A12
DR-09	Video access and downloadable materials shall be restricted according to enrollment and instructor settings (downloadable vs. view-only).	Content ownership / intellectual property	FR-S14, FR-I17
DR-10	Paid enrollments shall generate an invoice/receipt showing applicable taxes (e.g., GST).	Indian tax rules for digital services	FR-PG01–PG04

DR-11	Refund eligibility and timelines shall follow a published refund policy shown before purchase.	Consumer protection norms	FR-A24, FR-PG05
DR-12	Card details shall never be stored by the platform; only gateway-issued tokens may be kept.	RBI card-data storage rules; PCI-DSS	FR-PG09–PG10, NFR-15
DR-13	Course video shall be accessible: captions/transcripts and screen-reader/keyboard-friendly pages.	Inclusive education norms; WCAG	NFR-21
DR-14	Course content and progress data should be structured so they can be exported or integrated using common e-learning standards (e.g., SCORM/xAPI) where feasible.	E-learning interoperability standards	FR-I06, FR-S16
DR-15	Students shall be able to request export or deletion of their personal data, and the platform shall respond within the legally defined period.	DPDP Act / GDPR data-subject rights	NFR-31

Education and Assessment Domain:

ID	Domain Requirement	Basis (Domain Source)	Nature	Related FR / NFR
DR-ED 01	Certificates shall be issued only when the instructor-defined completion criteria are met. Each certificate shall carry a unique certificate ID, learner name, course title, issue date and issuer, and shall be verifiable through a public verification link or QR code.	Certification practice in online education (verifiability of credentials)	Domain rule	FR-S77, FR-S78, FR-S79, FR-I56, FR-I57, FR-A29
DR-ED 02	Certificates and course pages shall describe a certificate as a platform completion certificate. They shall not state or imply accreditation, degree or credit equivalence unless the course is offered under the approval of the relevant recognised body (e.g., UGC/AICTE-approved institution).	Consumer protection law on misleading claims; UGC/AICTE norms for degree and credit claims	Legal	FR-S07, FR-S78, FR-I05, FR-A06
DR-ED 03	Marking rules shall be applied exactly as configured (negative marking, partial marks for multiple-select, numerical tolerance) and shown to students before the attempt. The total marks of an assessment shall equal the sum of its question marks.	Transparent and reproducible grading	Domain rule	FR-S19, FR-S20, FR-S25, FR-I20, FR-I21, FR-I22
DR-ED 04	Published results are academic records. Any change (regrade, override, invalidation) shall create a new version with reason, actor and timestamp, and the earlier value shall be kept.	Academic record integrity	Domain rule	FR-I32, FR-I37, FR-S45, FR-A16, NFR-19
DR-ED 05	When a question is corrected or invalidated, one configured policy (e.g., award full marks, or drop the question and rescale) shall be applied uniformly to every affected attempt.	Fair assessment practice	Domain rule	FR-S45, FR-I36, FR-I37
DR-ED 06	Attempt limits, availability windows and time limits shall be enforced by the server. Client-side clocks shall not be trusted.	Assessment integrity	Standard	FR-S22, FR-S24, FR-S35, FR-S37, FR-I20, NFR-03, NFR-11
DR-ED 07	Prerequisites and eligibility rules set by the instructor shall be enforced at enrolment and at exam start, with a clear reason shown when access is denied.	Academic eligibility rules	Domain rule	FR-S07, FR-S08, FR-S34, FR-S35, FR-I11
DR-ED 08	All schedules shall be stored in UTC and shown in the user's local time (default IST), with the time zone displayed for exams and deadlines.	Learners in different time zones; fair exam timing	Standard	FR-S33, FR-S34, FR-I18, FR-I25

DR-ED 09	Assessment results, certificates and completion records shall be kept for a defined retention period so that certificates stay verifiable, subject to the user's erasure rights (DR-PR06).	Credential verification needs; data protection law	Domain rule	FR-S48, FR-S79, FR-I57, NFR-31, NFR-34
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Payments and Finance Domain (India):

ID	Domain Requirement	Basis (Domain Source)	Nature	Related FR / NFR
DR-PY 01	Payments shall be processed only through an RBI-authorised payment aggregator or gateway. The platform shall not store, process or log card number, CVV or expiry; it may keep only gateway-issued tokens and the last four digits.	RBI rules on card data storage and tokenisation; PCI DSS	Legal	FR-PG01, FR-PG09, FR-PG10, NFR-15
DR-PY 02	Saved cards and recurring or subscription payments shall need explicit customer consent and use tokens. Recurring debits shall follow the RBI e-mandate framework: advance notice before each debit, additional authentication and easy cancellation.	RBI e-mandate framework for recurring payments	Legal	FR-PG01, FR-PG10, NFR-15
DR-PY 03	The total payable amount (price, discounts, taxes, fees) shall be shown before payment. There shall be no hidden charges, pre-ticked add-ons or misleading urgency messages.	Consumer Protection Act 2019; CCPA Guidelines on Dark Patterns 2023	Legal	FR-S08, FR-I10
DR-PY 04	The refund and cancellation policy shall be shown before purchase. Refunds shall be credited to the original payment source within the timeline set by the policy and by gateway/RBI rules.	Consumer Protection (E-Commerce) Rules 2020; RBI rules	Legal	FR-PG05, FR-PG06, FR-A24, NFR-12
DR-PY 05	If money is debited but enrollment is not confirmed, the platform shall reverse or reconcile it within the turnaround time RBI prescribes for failed transactions. Enrolment shall be created only after a confirmed success (FR-PG04).	RBI turnaround time framework for failed transactions	Legal	FR-PG03, FR-PG04, FR-PG12, NFR-12
DR-PY 06	Every paid transaction (and the platform commission) shall have a GST-compliant tax invoice, with configurable tax rates and exemptions. Statutory deductions on instructor payouts (e.g., TDS) shall be applied where applicable, with statements provided.	GST law; Income Tax Act	Legal	FR-I10, FR-I48, FR-I49, FR-A25, NFR-33

DR-PY 07	Payouts shall go only to verified bank accounts in the instructor's own name. Bank account and required tax/identity details shall be verified before the first payout.	Anti-money-laundering and fraud-prevention practice	Legal	FR-I50, FR-PG07, FR-A26
DR-PY 08	Every transaction, refund and payout shall be recorded in an append-only ledger with the gateway reference. Duplicate webhooks or retries shall not create duplicate enrolments, refunds or payouts. The ledger shall be reconciled daily with gateway settlement reports.	Financial control practice	Standard	FR-PG12, FR-PG14, FR-PG15, FR-A23, NFR-16, NFR-19
DR-PY 09	For chargebacks and disputes, evidence (enrollment, access logs, accepted terms) shall be kept for the dispute period, and payouts for disputed transactions may be held.	Card network dispute process	Domain rule	FR-PG13, FR-A26
DR-PY 10	Purchases by users under 18 shall be made only through a guardian-authorised payment.	Indian Contract Act 1872 (minors cannot contract)	Legal	FR-S08, FR-PG01, DR-PR04
DR-PY 11	International payments shall be accepted only through a gateway authorised for cross-border payments, with currency conversion shown before payment.	FEMA and RBI rules on cross-border payments	Legal	FR-PG01, FR-PG02